

Microbiological Methods

A Polymerase Chain Reaction (PCR) Method to Detect Emerging Multidrug-Resistant *Salmonella* Infantis Harboring the pESI Plasmid in Seafood

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Abstract

Background: *Salmonella* Infantis is an emerging multidrug-resistant pathogen worldwide due to the acquisition of a megaplasmid, plasmid of emerging *Salmonella* Infantis (pESI). Reported initially in poultry, the distribution of pESI-harboring *S. Infantis* in other food types, including seafood, is unknown.

Objective: This study aimed to develop and optimize a PCR assay for detecting the pESI in *Salmonella* and non-*Salmonella* Enterobacteriales.

Methods: A duplex PCR targeting the *hlyA* gene and a pESI-associated gene of *S. Infantis* was designed, and the PCR conditions were optimized. The specificity and sensitivity of the assay were established using 119 *Salmonella* serovars and 51 non-*Salmonella* bacterial strains.

Results: All *Salmonella* isolates yielded *hlyA* PCR product, while only pESI *S. Infantis* was positive for both *hlyA* and pESI genes. No amplification product was obtained with the DNA of 51 non-*Salmonella* bacterial strains. The detection limit of the duplex PCR was 10⁴ CFU/mL of pure culture of pESI *S. Infantis*. The sensitivity of detection in artificially spiked shrimp meat was 1 CFU/g after 6 h of enrichment in lactose broth, followed by 12 h of selective enrichment in the Rappaport-Vassiliadis medium.

Conclusion: The duplex assay will help screen seafood for *Salmonella* in general and pESI *S. Infantis* in particular. Given its high sensitivity, the PCR will be a valuable tool for seafood quality assurance. This approach decreases the typical 3–6 day identification time of *Salmonella* to less than 24 h.

Highlights: *S. Infantis* carrying the highly transmissible megaplasmid (pESI) is a significant food safety concern. Given its rapid geographical spread and high antimicrobial-resistant traits, it is necessary to have a molecular tool that detects pESI-harboring *Salmonella*. This study successfully developed a duplex PCR assay that simultaneously detects *Salmonella enterica* and pESI *S. Infantis*. This molecular tool will help understand the distribution, sources, and spread of the multidrug-resistance (MDR) plasmid in the food environment.

Non-typhoidal *Salmonella* (NTS), comprising over 2600 serovars, is the most diverse human pathogen with a vast host range (1). Although all *Salmonella* serovars are considered potentially pathogenic, only a few serovars contribute to most food-borne outbreaks (2,3). Foods of animal origin, such as poultry, eggs, meat, dairy products, and seafood, are more commonly implicated in NTS-associated gastroenteritis (4,5). The NTS serovars are increasingly becoming resistant to antibiotics commonly employed to treat gastroenteritis, such as the extended-spectrum cephalosporins and fluoroquinolones (6,7). *Salmonella enterica* serovar Infantis is one of the rapidly spreading NTS globally, posing a significant threat to public health (1, 7). *Salmonella enterica* subsp. *enterica* serovar Infantis (*S. Infantis*) is one of the NTS that is spreading globally and poses a significant threat to public health (1, 7). The rapid emergence of the *S. Infantis* clone in Israel was reported for the first time because of the acquisition of a megaplasmid, termed pESI (plasmid of emerging *Salmonella*

Infantis; 8). This conjugative plasmid confers resistance to multiple antimicrobial drugs including tetracyclines, ampicillin, cephalosporins (ceftazidime and ceftriaxone), sulfamethoxazole-trimethoprim, sulfonamides, chloramphenicol, aminoglycosides (streptomycin and kanamycin), quinolones (nalidixic acid and ciprofloxacin), heavy metals (through mercury- and arsenic-resistant genes), and disinfectants, and enhances virulence-associated phenotypes and the pathogenicity of its bacterial host (4,5, 8–10). A recent study reported the incidence of *S. Infantis* in America (42.9%), Europe (29.8%), Western Pacific (17.2%), Eastern Mediterranean (6.6%), Africa (3.4%), and Southeast Asia (0.1%) with identified sources mainly due to contaminated chicken and other foods (11). A recent study from India reported the isolation of the emerging strain of non-typhoidal *S. Infantis* carrying the mega plasmid pESI (285 kb) and resistant to multiple drugs from marine shrimp (12). The pESI plasmid has been identified in four serovars, namely *S. Muenchen*, *S. Schwarzengrund*, *S. Agona*,

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and S. Senftenberg, mostly from poultry (broiler and chicken), sources (7). Given its rapid dissemination and food safety concern, studies are necessary to understand the prevalence and the sources of *S. Infantis* harboring the pESI plasmid.

In this study, a duplex PCR was developed for the detection of *S. Infantis* harboring the pESI plasmid in seafood by detecting more than two loci in a single reaction simultaneously, which enables the detection of all *Salmonella* as well as pESI-carrying *Salmonella* in a single reaction.

Experimental

Salmonella isolates (n = 130) used in this study belonged to 13 serotypes (Table 1). All isolates were from seafood, isolated previously using the standard protocol outlined in the Bacteriological Analytical Manual of the U.S. Food and Drug Administration (13,14). Molecular confirmation of the isolates was made by *invA* gene-specific PCR (15). The isolates were serotyped at the National *Salmonella* and *Escherichia* Centre, CRI-Kasauli, HP, India. For all bacteriological and molecular studies in this work, *Salmonella enterica* subsp. *enterica* serovar *Infantis* strain JS5, which carries the pESI megaplasmid (285 kb), was used as the reference strain (12).

Revival and Purification of *Salmonella* Stock Isolates

This study used 119 isolates of *Salmonella* representing several serovars to assess the specificity of duplex PCR (12, 16). The bacterial strains stored in glycerol broth were revived in Luria Bertani (LB) broth and streaked on LB agar to obtain pure colonies. Each *Salmonella* culture was streaked on xylose lysine deoxycholate (XLD) agar plates and incubated at 37°C overnight to confirm the colony phenotypes. Further confirmation of *Salmonella* spp. was made by *invA* gene-specific PCR (17).

Selection of Oligonucleotide Primers

Two sets of oligonucleotide primers were selected to detect *Salmonella enterica* and *Salmonella Infantis* carrying pESI plasmid simultaneously. Accordingly, the *hliA* gene present in all *Salmonella* serovars was targeted to detect *S. enterica* (18). The *hliA*-specific forward (CTGCCGAGTGTAAAGGAT) and reverse (CTGTGCGCTATAATCGCATGT) amplifying a 497 region were

selected from a previous study (18). The forward (AACCCACAGT ACTGCCAC) and reverse (GCTGGCATCTCAATGGTG) primers targeting the pESI plasmid were designed from our previously published whole genome sequence of *S. Infantis* JS5 (accession no. PRJNA794263; 12), to amplify a 287 bp portion of the pESI megaplasmid. The Primer-BLAST program was used to design gene-specific primers (19).

Optimization of Duplex PCR Conditions

A gradient from 55 to 62°C was selected, since *hliA* primers amplify optimally at 60°C (18), and pESI primers yielded the expected amplification product at an annealing temperature range of 58 to 63°C. Duplex PCR was optimized in a 25 µL reaction mixture consisting of 12.5 µL 2X Emerald Amp GT PCR Master Mix (Takara Inc., Japan), 2 µL each of forward and reverse primers, and 2 µL of DNA from *S. Infantis* JS5 carrying pESI megaplasmid; sterile deionized water was used as a negative control in all PCR assays. PCR products (10 µL) were analyzed on ethidium bromide-stained 2% agarose gel and visualized under a UV transilluminator.

Specificity of Oligonucleotide Primers

The specificity of the designed duplex PCR was determined in this study using 119 *Salmonella* isolates representing different serovars and 51 non-*Salmonella* bacteria comprising Enterobacterales and other Gram-negative and -positive bacteria.

Determination of Duplex PCR Sensitivity Using Broth Culture of *S. Infantis*

The growth curves of ATCC 14028 (*S. Typhimurium*) and *S. Infantis* JS5 were constructed using the spectrophotometric method. *S. Infantis* JS5 was grown exponentially to the mid-log phase (OD₆₂₀ = 0.47 and cell count = 10⁹ CFU/mL) in LB broth and 10-fold serially diluted. The cell count was determined by viable plate count on XLD agar plates. The broth culture of JS5 was serially diluted using fresh LB broth in 1.5 mL microcentrifuge tubes to obtain 10⁸ to 1 CFU/mL. Two microliters of broth culture lysate from each dilution were used as the template for PCR.

Table 1. Results of specificity testing of duplex PCR using diverse *Salmonella* serovars

Serotype (No. of isolates)	Isolate number	Duplex PCR result	
		<i>hliA</i>	pESI
<i>S. Alachua</i> (3)	195, 295, 296	+	–
<i>S. Richmond</i> (3)	JS13, JS15, JS17	+	–
<i>S. Newport</i> (9)	361, 800, 806, 824, 1042, 1043, 1061, 1064, 1065	+	–
<i>S. Bareilly</i> (67)	352, 354, 356, 683, 684, 685, 686, 687, 691, 692, 693, 694, 695, 696, 697, 698, 699, 700, 701, 702, 703, 704, 705, 706, 707, 708, 709, 710, 712, 713, 714, 826, 925, JS6, JS7, JS8, JS9, JS10, JS11, JS12, JS14, JS16, JS18, JS19, JS20, JS21, JS22, JS23, JS24, JS25, JS26, JS27, JS28, JS29, JS30, JS31, JS32, JS33	+	–
<i>S. Kentucky</i> (8)	298, 299, 300, 307, 318, 362, 365, 1066	+	–
<i>S. Haifa</i> (3)	JS1, JS3, JS4	+	–
<i>S. Volta</i> (2)	506, 555	+	–
<i>S. Paratyphi B</i> (8)	316, 323, 327, 328, 335, 360, 363, 427	+	–
<i>S. Typhimurium</i> (15)	313, 314, 315, 317, 324, 326, 329, 330, 331, 332, 333, 1041, 1044, JS2	+	–
<i>S. Tennessee</i> (7)	343, 346, 355, 357, 369, 922, 1063	+	–
<i>S. Lindenburg</i> (1)	342	+	–
<i>S. Infantis</i> (1)	JS5	+	+
<i>S. Weltevreden</i> (2)	503, 504	+	–

Determination of Duplex PCR Sensitivity in Shrimp Spiked with *S. Infantis* and Without Enrichment

For the spiking experiment, *S. Infantis* JS5 was grown in LB broth to an OD₆₂₀ of 0.41 and serially diluted using fresh LB broth in 1.5 mL microcentrifuge tubes to obtain counts of 10⁸ to 1 CFU/mL. One milliliter from the dilutions representing 10⁵ to 10² CFU/mL was used to spike 10 g of individual shrimp (*Penaeus vannamei*) tissue homogenate in physiological saline. Spiking resulted in a 10-fold dilution of bacterial numbers. The viable bacterial count in the homogenate was determined by 10-fold serial dilution of the homogenate in physiological saline, followed by plating on XLD agar in duplicates. For DNA extraction, the homogenate was centrifuged at 800 revolutions per min (rpm) for 5 min to pellet larger meat particles, and the supernatant was centrifuged at 10000 rpm for 10 min to pellet bacterial cells. The cell pellet was subjected to DNA extraction using a Wizard DNA extraction kit (Promega, WI, USA) according to the manufacturer's instructions. The duplex PCR was performed on the extracted DNA. All shrimp samples used in artificial spiking experiments were screened for *Salmonella* by *invA* PCR and confirmed that the samples were *Salmonella*-negative.

Sensitivity of Duplex PCR in *Salmonella*-Spiked Shrimp after Pre- and Selective Enrichment

S. Infantis JS5 was grown to an OD₆₂₀ of 0.41 (cell count = 10⁸ CFU/mL) and serially diluted. One milliliter of each of the serially diluted JS5 cultures consisting of 10⁴ to 10² cells (CFU/mL) was added to 10 g of individual *P. vannamei* tissue homogenates and pre-enriched in 90 mL lactose broth at 37°C for 6, 12, 18, and 24 h. Following pre-enrichment, 1 mL of each sample was subjected to DNA extraction. For selective enrichment, 0.1 mL from the lactose broth culture was inoculated into 10 mL Rappaport-Vassiliadis (RV) broth and incubated at 42.0 ± 0.5°C in a water bath for 24 h. Following selective enrichment in RV medium, 1 mL of each sample was subjected to DNA purification. The RV broth cultures were streaked on XLD agar and Hektoen Enteric Agar (HEA) plates to ensure the presence of *S. Infantis*. The experiment was repeated by performing selective enrichment for 6, 12, 18, and 24 h at 42 ± 0.5°C. At each interval, 1 mL of each sample was subjected to DNA purification.

Results and Discussion

Development and Optimization of Duplex PCR

A duplex PCR was designed to target the *hilA* gene of all *Salmonella* serovars and the pESI plasmid-associated *S. Infantis* harboring the pESI plasmid. Multiplex PCR techniques have been developed to identify and distinguish significant *Salmonella* serovars with high accuracy (2, 20–23). However, there is no PCR method to specifically detect the pESI-harboring *Salmonella*. Since NTS are often reported from seafood in India (15, 24), it is essential to have a PCR method to specifically identify this rapidly spreading *S. Infantis* carrying the pESI plasmid. The modified duplex PCR generated two amplicons: a 497 bp *hilA*-specific band and a 287 bp pESI plasmid-specific band. In duplex PCR, the best amplification of two genes was obtained at an annealing temperature of 60°C, suggesting that selected primers were sufficiently specific for detecting pESI plasmid-harboring *S. Infantis*. The result showed clear and distinct amplification products for both *hilA* (497 bp) and pESI plasmid-specific genes (287 bp). The optimized PCR conditions consisted of an initial denaturation of 5 min at 94°C, followed by 30 cycles of 30-second denaturation at 94°C, 30-second annealing at 60°C, and 30-second extension at 72°C. The final extension was

5 min at 72°C. The *hilA* primers could detect almost all *Salmonella* serovars without any nonspecific product in *Salmonella*-related strains (25). Both *invA* and *hilA* genes are marker genes of *S. enterica* with potential diagnostic uses (17,18, 26).

Determination of Specificity

Non-*Salmonella* species, including multidrug-resistant *Escherichia coli* from seafood, did not yield any amplification product with the duplex PCR, suggesting that the assay was highly specific. The duplex PCR was applied to all serotyped laboratory stock cultures of *Salmonella enterica*. All the laboratory isolates of *Salmonella* were confirmed by *invA*-specific PCR. All stock isolates, including 119 known serotypes and the isolates of *Salmonella* (27), were subjected to the newly developed duplex PCR assay. All *Salmonella* isolates showed amplification of the *hilA* gene, confirming their identity as *Salmonella* (Table 1). However, none of the isolates harbored the pESI plasmid and were uniformly negative for the pESI-associated gene targeted in the duplex PCR (Table 2). This unequivocally establishes the specificity of the PCR and the fact that no nonspecific amplifications were obtained with any of the isolates tested in this study further validates the stringency of PCR conditions established in this study.

Sensitivity of Detection of Duplex PCR Assay

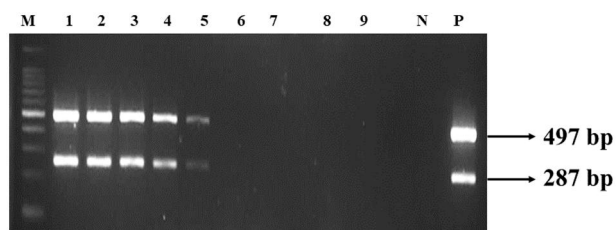
The growth curve of *S. Infantis* JS5 was derived by plotting OD₆₂₀ values and log CFU/mL obtained at 4-hour intervals against the time. The growth parameters were derived from the growth curve of *S. Infantis* JS5. The specific growth rate constant (μ) and the generation time (g) were determined as 1.8444 h⁻¹ and 22.5 min, respectively. First, the sensitivity of the assay was determined using known numbers (CFU/mL) of *S. Infantis* grown in LB broth and serially diluted to obtain counts from 10 to 10⁸ CFU/mL. The duplex PCR was able to detect 10⁴ CFU/mL of pure culture of *S. Infantis* JS5 (Figure 1). In contrast, no amplification signal was obtained in shrimp artificially spiked with 10⁴ CFU/g *S. Infantis*, suggesting an inhibitory effect of shrimp matrix on PCR amplification (data not shown). Studies have reported matrix-dependent varying sensitivity in the detection of *Salmonella*. In order to ensure reliable PCR amplification, a sensitivity of 10³ CFU per mL of sample (one cell per PCR reaction tube) has been reported to be necessary (28). Another study found that the detection limit was 100 pg with genomic DNA and 3 × 10⁴ CFU/mL with *S. Typhimurium* (25). Similarly, in artificially spiked chicken feces samples, the LOD was reported to be 10⁶ CFU/g (27).

Sensitivity of Duplex PCR in JS5 Spiked Shrimp after Enrichment

Shrimp (*P. indicus*) was spiked with *S. Infantis* JS5 at 10, 10², and 10³ CFU/g and subjected to the standard pre-enrichment and selective enrichment procedures, followed by duplex PCR. A study from India reported that the PCR detection of *S. enterica* from crude lysates prepared from two enrichment broths, such as selenite cysteine broth and tetrathionate broth, was more efficient than the culture method in seafood (29). Past studies reported that RV broth is the most efficient *Salmonella*-selective medium for seafood (14, 30). Table 3 shows the PCR results with artificially spiked shrimp. The first detection signal from lactose broth was obtained after 6 and 12 h of pre-enrichment of shrimp inoculated with 10³ CFU/g and 10² CFU/g JS5, respectively (Figure 2). In the case of shrimp spiked with 10 CFU/g JS5, no amplification signal could be detected after 24 h of enrichment in lactose broth. However, all RV broth tubes, including those inoculated from PCR-negative lactose broths, showed PCR amplifications after

Table 2. Non-*Salmonella* bacterial strains used in this study and the results of duplex PCR

Bacterial strain	Isolate number	Duplex PCR results	
		hilA	pESI
<i>Escherichia coli</i>	Ec114, ECJ53, KAM32, A5, ETEC, USFDA O157: H7, Ec33, Ec34, Ec35, Ec36, Ec40, Ec41, Ec42, Ec43, Ec44, Ec45, Ec46, Ec47, Ec48, Ec49, Ec50, Ec51	—	—
<i>Bacillus subtilis</i>	—	—	—
<i>Aeromonas hydrophila</i>	AH1	—	—
<i>Arcobacter butzleri</i>	ATCC 49616	—	—
<i>Enterobacter cloacae</i>	BMS37	—	—
<i>Enterococcus casseliflavus</i>	JMS41	—	—
<i>Vibrio parahaemolyticus</i>	O3: K6, Niced (tdh ⁺)	—	—
Methicillin-resistant <i>Staphylococcus aureus</i>	ATCC 25923	—	—
<i>B. cereus</i>	JML33	—	—
<i>Klebsiella pneumoniae</i>	JMY4, BHC1, KD104, KD53, KD121, JML58, BMS60, BMV56, JMY4, BHC1, JSL21	—	—
<i>Proteus mirabilis</i>	BML42, BML43, JSY14, JMS48, JMS68, JMS69, BMS68, BMS70, BMS67, BMS74, JMS75, JML66	—	—
<i>Serratia marcescens</i>	BSS23, JSY13, JMS50	—	—
<i>Pseudomonas fluorescens</i>	MTCC 667	—	—
<i>Cronobacter sakazakii</i>	ATCC 29544	—	—
<i>Enterobacter hormaechei</i>	BML46	—	—
<i>Raoultella ornithinolytica</i>	BML25	—	—
<i>Staphylococcus aureus</i>	ATCC BAA 976	—	—
<i>Bacillus subtilis</i>	BML32	—	—
<i>V. cholerae</i>	O395	—	—

**Figure 1.** Sensitivity of detection of broth-grown *S. Infantis* JS5 by duplex PCR. Lane M: 100 bp ladder (Takara Bio, Japan); Lane N: negative control; Lane P: positive control (*S. Infantis* JS5 DNA); Lanes 1–9: 10^8 , 10^7 , 10^6 , 10^5 , 10^4 , 10^3 , 10^2 , 10, and 1 CFU/mL, respectively.

24 h (Figure 2), emphasizing the importance of selective enrichment in detecting *Salmonella* in shrimp matrix.

In the subsequent experiment, we determined the duration of selective enrichment in RV necessary to detect *S. Infantis* in shrimp inoculated with 10 and 1 CFU/g. In this experiment, pre-enrichment in lactose broth was performed for 6, 12, 18, and 24 h. After each incubation period, RV broth was inoculated and incubated for 6, 12, 18, and 24 h before DNA extraction and PCR (Table 4). As can be seen from Figure 3, pre-enrichment in lactose broth alone could not detect *S. Infantis* even after 24 h. In the case of shrimp spiked with 10 CFU/g JS5, the first detection signal by PCR was obtained in RV broth inoculated with lactose broth incubated for 12 h. In the case of shrimp spiked with 1 CFU/g JS5, the first detection signal was obtained in RV broth inoculated with lactose broth incubated for 18 h. In both cases, the positive RV broth tubes were incubated for 6 h only. In other words, 12 h of pre-enrichment in lactose broth followed by 6 h of selective enrichment in RV broth could detect 10 CFU/g *S. Infantis* pESI.

In contrast, in the case of 1 CFU/g, 18 h of pre-enrichment and 6 h of selective enrichment were required to detect the bacterium (Table 4). These results suggest that the duplex PCR developed in this study has sufficiently high sensitivity in a seafood matrix, in view of the fact that *Salmonella* is a zero-tolerant pathogen in

Table 3. Detection of *S. Infantis* pESI in spiked shrimp by duplex PCR after different enrichment periods

No. of bacteria inoculated, CFU/g	Results of duplex PCR								
	Lactose broth ^a					RV broth ^b			
	0 h	6 h	12 h	18 h	24 h	6 h	12 h	18 h	24 h
10 ³	—	+	+	+	+	+	+	+	+
10 ²	—	—	+	+	+	+	+	+	+
10	—	—	—	—	—	+	+	+	+
1	—	—	—	—	—	+	+	+	+

^a Direct detection of *S. Infantis* pESI in lactose broth at different intervals during 24 h of pre-enrichment period.

^b The time indicated corresponds to the duration of pre-enrichment in lactose broth before inoculation into RV broth. All RV broth tubes were incubated for 24 h.

seafood. This high sensitivity is necessary to detect multidrug-resistant pathogens such as *S. Infantis* pESI in shellfish. A few published studies have highlighted the importance of pre- and selective enrichment in the sensitive detection of *Salmonella* in foods. An 8-hour PCR test was developed for *Salmonella* detection in seafood by 284 bp *invA* with a detection limit of 2 CFU per 250 mL (13). Similarly, the threshold sensitivity was found to be 10^2 CFU/mL of pure culture of *Salmonella*, and the 18 h pre-enrichment step could detect as low as 3 CFU/25 g viable *Salmonella* in artificially contaminated food samples such as minced beef, whole egg, and fish (11). The enrichment step before PCR enables the detection of viable cells, as non-viable cells and PCR inhibitory substances are diluted during pre-enrichment and selective enrichment steps, leading to detectable multiplication of the target bacterium even when present in low numbers in the sample (31). Compared to the conventional culture method, PCR detection of *Salmonella* in chicken, clinical samples, and animals after enrichment significantly enhances detection efficiency (32). PCR detection after different enrichment periods for *Salmonella* has been reported by a few studies in poultry meat (6, 11, 33).

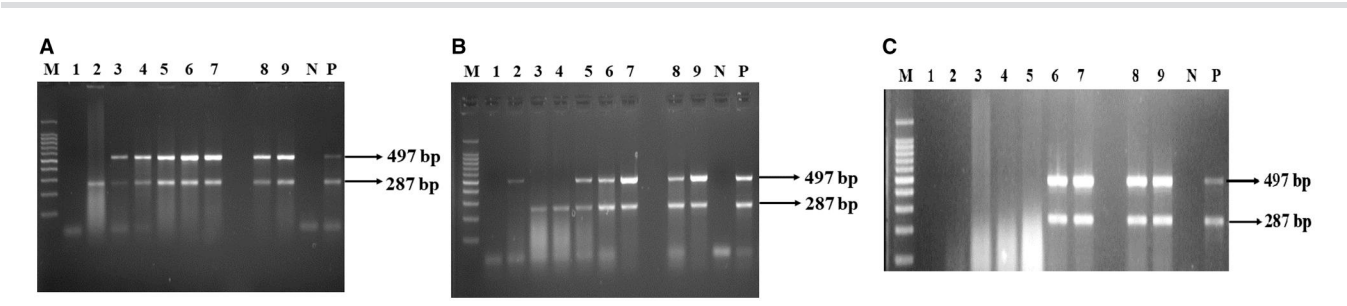


Figure 2. Sensitivity of detection in shrimp artificially spiked with *S. Infantis* JS5 at 10^3 CFU/g (A), 10^2 CFU/g (B), and 10 CFU/g (C) after enrichment. Lane M: 100 bp ladder (Takara Bio, Japan); Lanes 1–5: DNA extracted from lactose broth after 0, 6, 12, 18, and 24 h of pre-enrichment; Lanes 6–9: DNA extracted from RV broth inoculated from lactose broth corresponding to 6, 12, 18, and 24 h of pre-enrichment. In all these cases, RV broth was incubated for 24 h after inoculation from respective lactose broths; Lane N: negative control; Lane P: positive control (*S. Infantis* JS5 DNA).

Table 4. Sensitivity of detection of *S. Infantis* pESI in artificially spiked shrimp following enrichment for different durations

Spiking level, CFU/g	Duration of incubation															
	PCR results with pre-enrichment in lactose broth/selective enrichment in RV broth, h/h															
	6/6	12/6	18/6	24/6	6/12	12/12	18/12	24/12	6/18	12/18	18/18	24/18	6/24	12/24	18/24	24/24
10	–	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
1	–	–	+	+	+	+	+	+	+	+	+	+	+	+	+	+

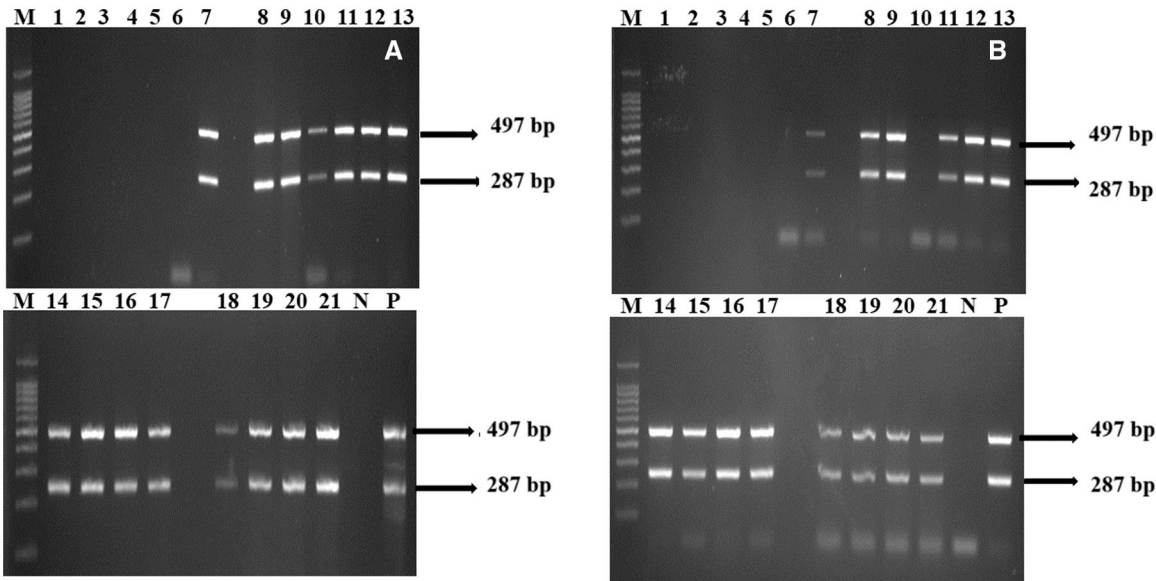


Figure 3. Sensitivity of detection of *S. Infantis* in shrimp spiked with 10 CFU/g (A) and 1 CFU/g (B). Lane M: 100 bp molecular marker (Takara); Lane N: negative control; Lane P: positive control; Lanes 1–5: lactose broth incubated for 6, 12, 18, and 24 h; Lanes 6–9: RV broth inoculated from 6 h incubated lactose broth, and enriched for 6, 12, 18, and 24 h; Lanes 10–13: RV broth inoculated from 12 h incubated lactose broth, and enriched for 6, 12, 18, and 24 h; Lanes 14–17: RV broth inoculated from 18 h incubated lactose broth, and enriched for 6, 12, 18 and 24 h; Lanes 18–21: RV broth inoculated from 24 h incubated lactose broth, and enriched for 6, 12, 18, and 24 h.

However, the LOD depends on the oligonucleotide primers, thermocycling conditions, and the food matrix itself. In the present study, a commercial, silica column-based method was used to extract DNA from enrichment samples, and this method was found to yield good results as a LOD of 1 CFU/g of shrimp could be achieved after pre- and selective enrichments.

Conclusions

This study offers an accelerated testing tool for seafood to detect pESI plasmid present in *Salmonella* *Infantis*, other *Salmonella* serovars, and other non-*Salmonella* bacterial strains. Considering its

high specificity and sensitivity, the duplex PCR assay could be used in seafood QA and *Salmonella* monitoring programs. This approach also decreases the typical 3–6 day identification time to less than a day. This effective method will be beneficial as a standard test for confirmation of pESI plasmid-carrying *S. Infantis* in clinical and environmental samples and the food chain.

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Conflict of Interest

The authors declare no conflict of interest.

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